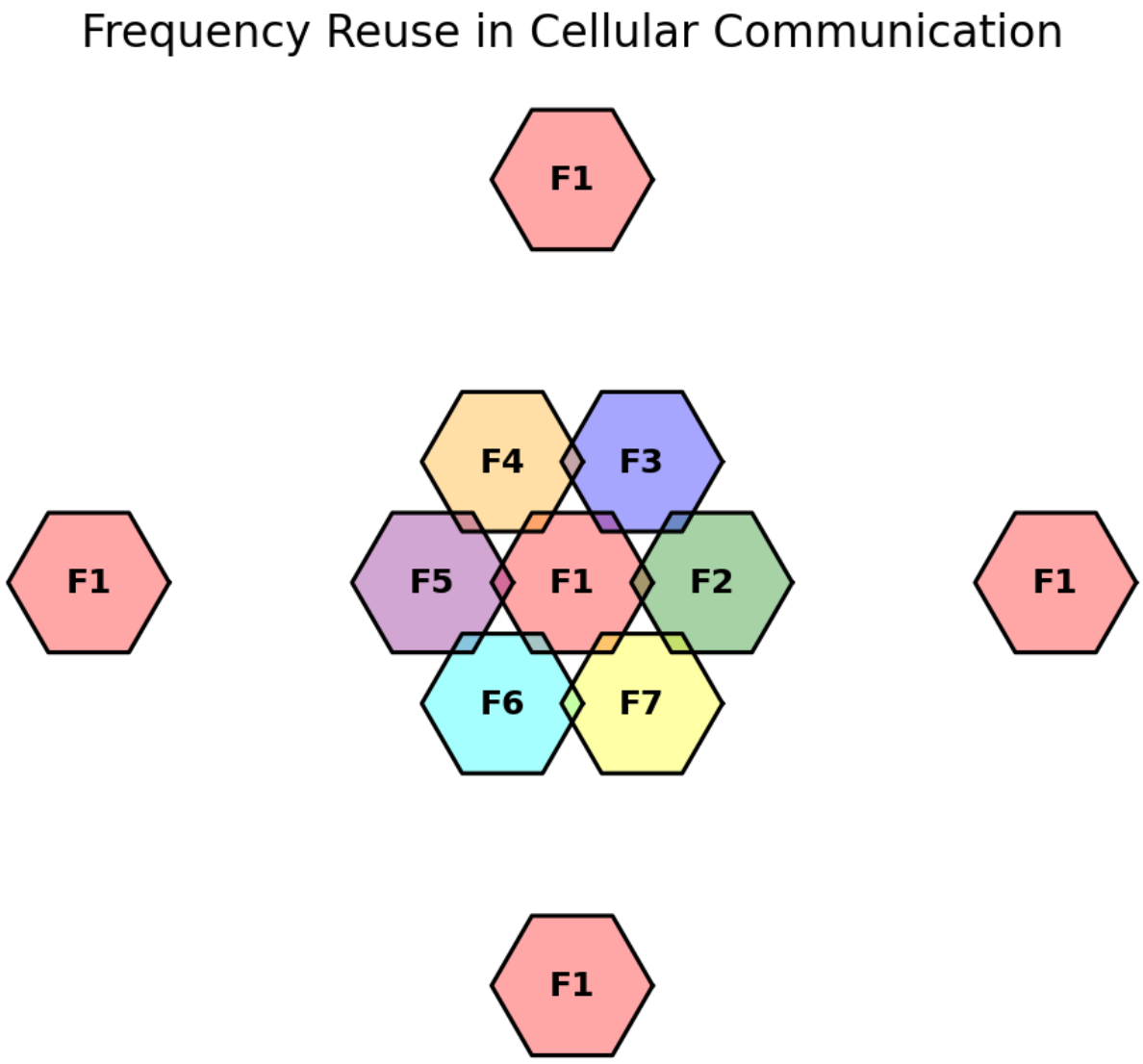
Wireless and Mobile Communication Lab — Experiment Outputs (3–10)

EXPERIMENT 3 — Frequency Reuse in Cellular Systems — OUTPUT



Observation Table:

Cell	Assigned Frequency
Cell 1	F1
Cell 2	F2
Cell 3	F3

Cell	Assigned Frequency
Cell 4	F4
Cell 5	F5
Cell 6	F6
Cell 7	F7

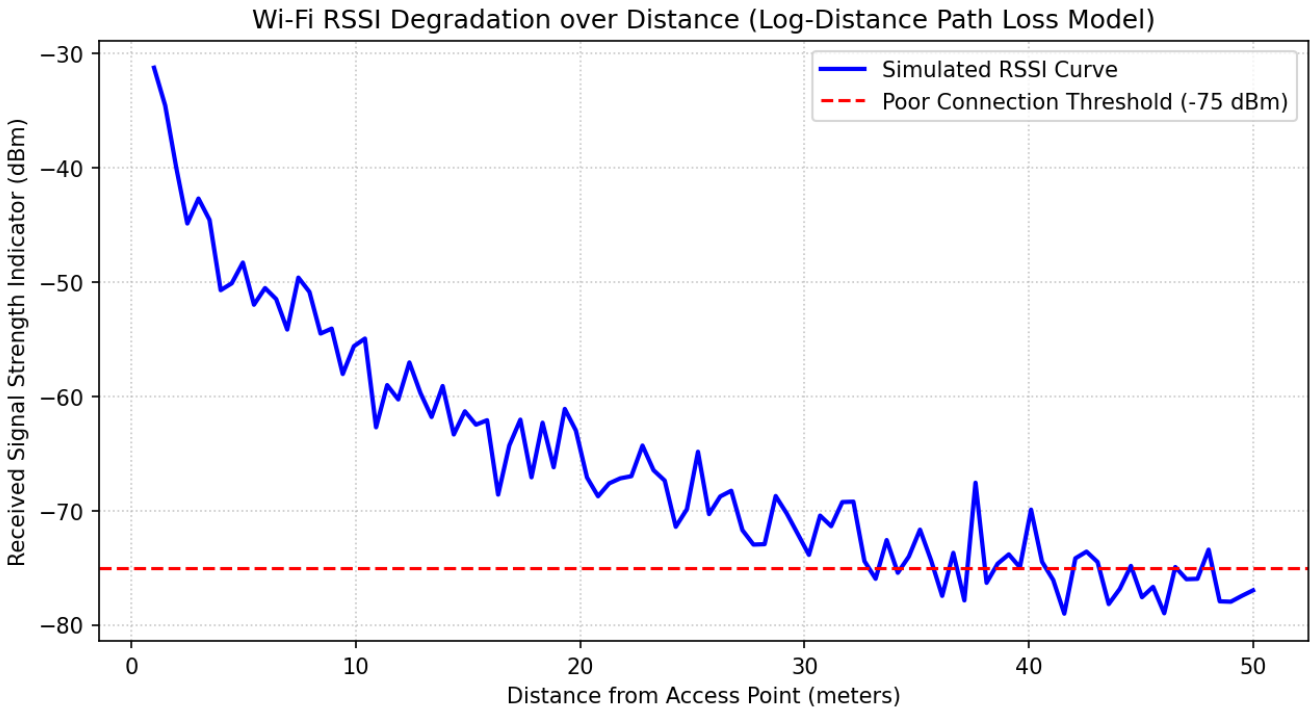
- Seven hexagonal cells with F1 through F7 allocation displayed.
- Reuse of F1 in four distant cells at coordinates (0,5), (0,−5), (6,0), (−6,0).

EXPERIMENT 4 — Wi-Fi SSID/RSSI Simulation & Channel Mapping — OUTPUT

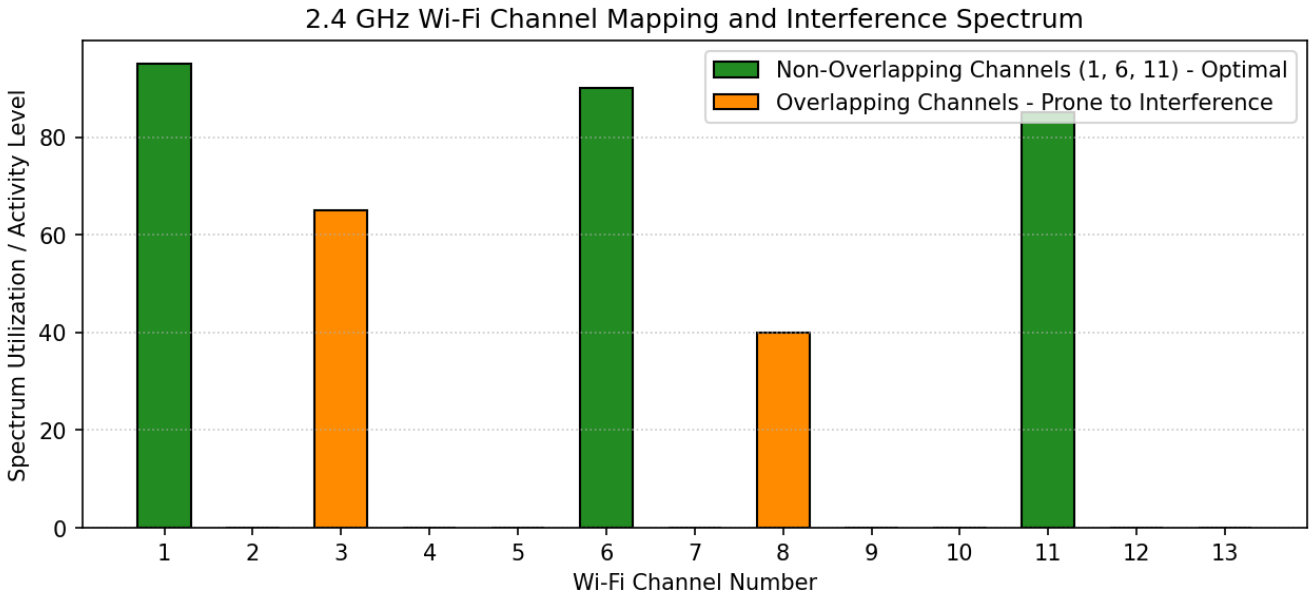
```
Google Colab Environment Initialized Successfully.
--- Table: Distance vs. RSSI Degradation ---
```

Distance (m)	Simulated RSSI (dBm)	Connection Status
1.0	-31.24	Excellent
10.4	-54.95	Good
20.3	-67.10	Good
30.2	-73.88	Fair
40.1	-69.92	Good
50.0	-76.98	Fair

Graph 1: RSSI Degradation over Distance



Graph 2: 2.4 GHz Channel Mapping and Interference Spectrum

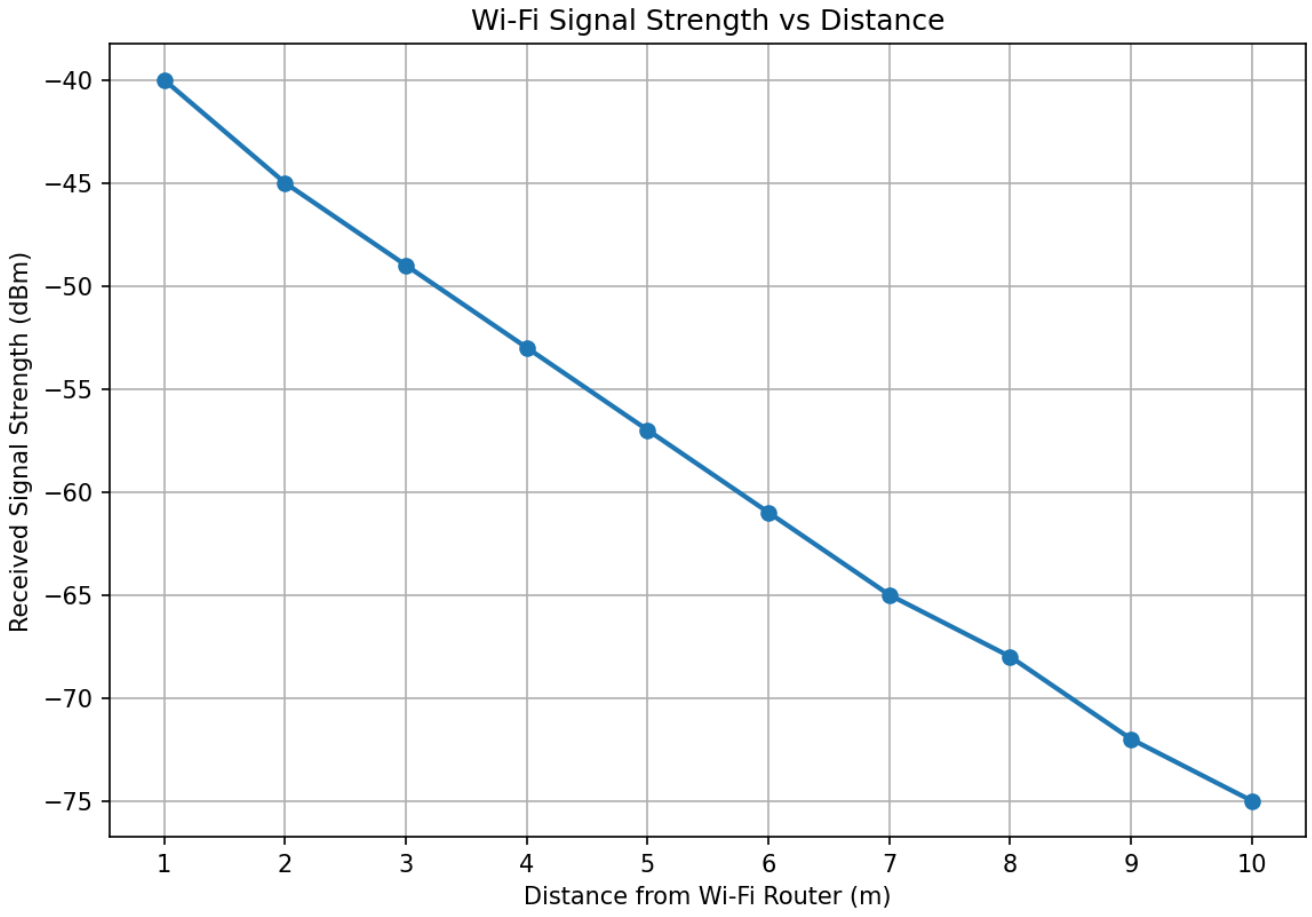

2.4 GHz Wi-Fi Channel Mapping and Interference Spectrum
Non-Overlapping Channels (1, 6, 11) - Optimal
Overlapping Channels - Prone to Interference
Spectrum Utilization / Activity Level
0
20
40
60
80
1
2
3
4
5
6
7
8
9
10
11
12
13
Wi-Fi Channel Number

EXPERIMENT 5 — Wi-Fi Signal Strength vs Distance — OUTPUT

SAMPLE DATA as provided in the manual

Distance (m)	RSSI (dBm)
1	-40
2	-45
3	-49
4	-53
5	-57
6	-61
7	-65
8	-68
9	-72
10	-75

Graph: Wi-Fi Signal Strength vs Distance

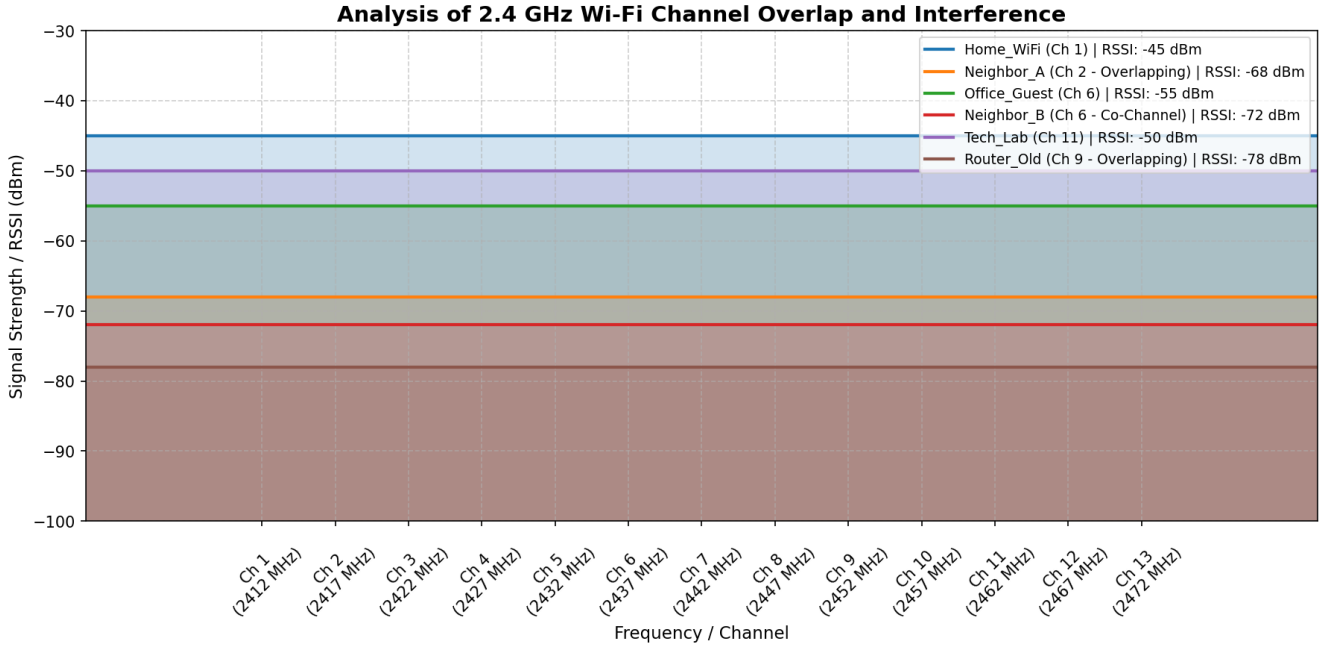


EXPERIMENT 6 — Wi-Fi Channel Interference Analysis — OUTPUT

Scanned Wi-Fi Networks:

SSID	Channel	RSSI (dBm)
Home_WiFi (Ch 1)	1	-45
Neighbor_A (Ch 2 - Overlapping)	2	-68
Office_Guest (Ch 6)	6	-55
Neighbor_B (Ch 6 - Co-Channel)	6	-72
Tech_Lab (Ch 11)	11	-50
Router_Old (Ch 9 - Overlapping)	9	-78

Graph: 2.4 GHz Wi-Fi Channel Overlap and Interference (2400–2484 MHz)



EXPERIMENT 7 — Network Throughput, Latency & Jitter Measurement — OUTPUT

```
============================================================
LATENCY AND PACKET LOSS MEASUREMENT
============================================================
Target Host : 8.8.8.8
Packets     : 15

Packet 01 : RTT = 102.00 ms
Packet 02 : RTT = 74.80 ms
Packet 03 : RTT = 26.50 ms
Packet 04 : RTT = 39.50 ms
Packet 05 : RTT = 34.40 ms
Packet 06 : RTT = 59.70 ms
Packet 07 : RTT = 31.20 ms
Packet 08 : RTT = 66.60 ms
Packet 09 : RTT = 43.10 ms
Packet 10 : RTT = 63.90 ms
Packet 11 : RTT = 31.10 ms
Packet 12 : RTT = 30.90 ms
Packet 13 : RTT = 66.00 ms
Packet 14 : RTT = 76.90 ms
Packet 15 : RTT = 26.50 ms

============================================================
THROUGHPUT MEASUREMENT
============================================================
Finding the best available Speedtest server...
Server : Gazon Communications India Ltd (Pune)

Measuring Download Throughput...
Download : 20.90 Mbps

Measuring Upload Throughput...
Upload   : 17.38 Mbps
Latency  : 44.58 ms

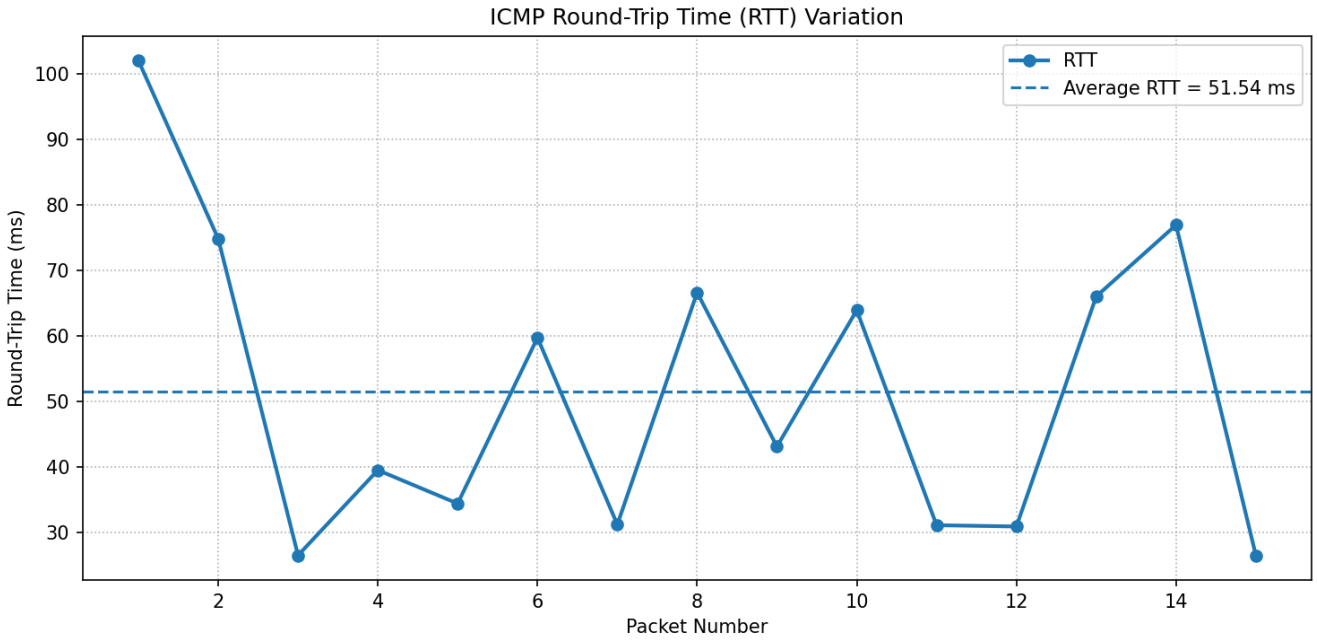
======================================================================
          NETWORK PERFORMANCE RESULTS
======================================================================
Target Host : 8.8.8.8
----------------------------------------------------------------------
Packets Sent                      : 15
Packets Received                  : 15
Packet Loss (%)                   : 0.00
Minimum RTT (ms)                  : 26.50
Maximum RTT (ms)                  : 102.00
Average RTT (ms)                  : 51.54
```

```
Median RTT (ms)                    : 43.10
RTT Standard Deviation (ms)        : 22.22
95th Percentile RTT (ms)           : 84.43
Average Jitter (ms)                : 25.46
Maximum Jitter (ms)                : 50.40
---------------------------------------------------------------------
Download Throughput (Mbps)         : 20.90
Upload Throughput (Mbps)           : 17.38
Speedtest Latency (ms)             : 44.58
Speedtest Server                   : Gazon Communications India Ltd (Pune)
=====================================================================
```

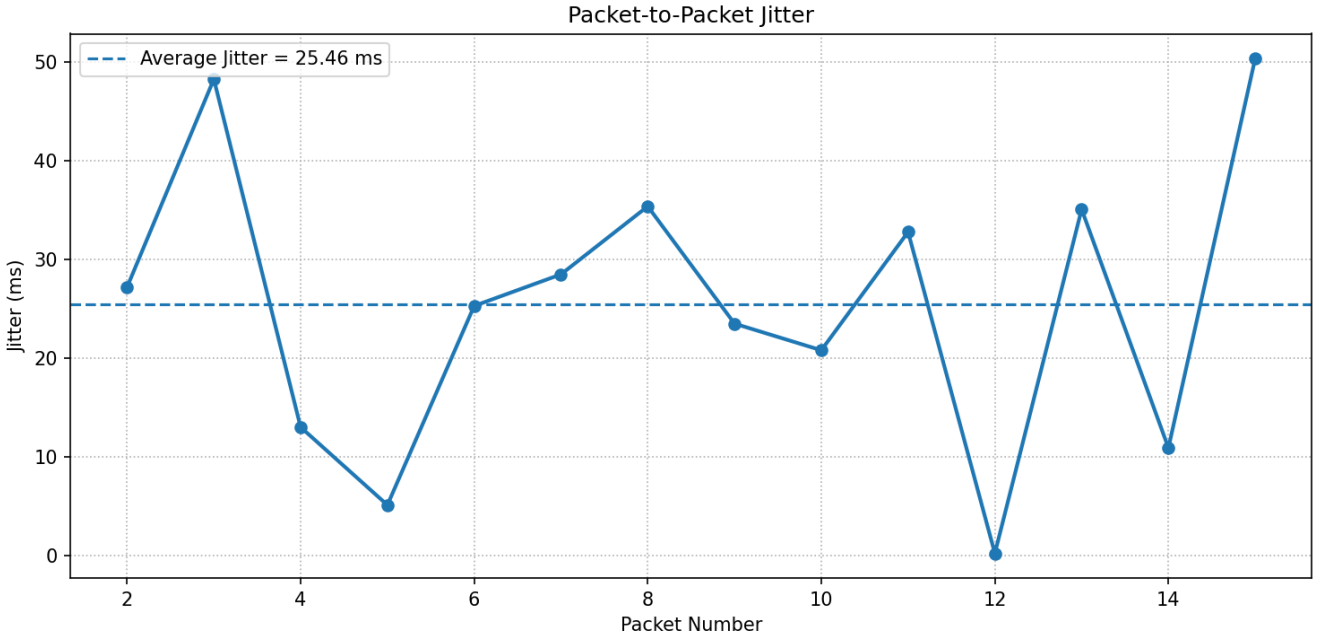
Observation Table:

Parameter	Measured Value
Packets Sent	15
Packets Received	15
Packet Loss	0.00 %
Minimum RTT	26.50 ms
Maximum RTT	102.00 ms
Average RTT	51.54 ms
Median RTT	43.10 ms
RTT Standard Deviation	22.22 ms
95th Percentile RTT	84.43 ms
Average Jitter	25.46 ms
Maximum Jitter	50.40 ms
Download Throughput	20.90 Mbps
Upload Throughput	17.38 Mbps
Speedtest Latency	44.58 ms

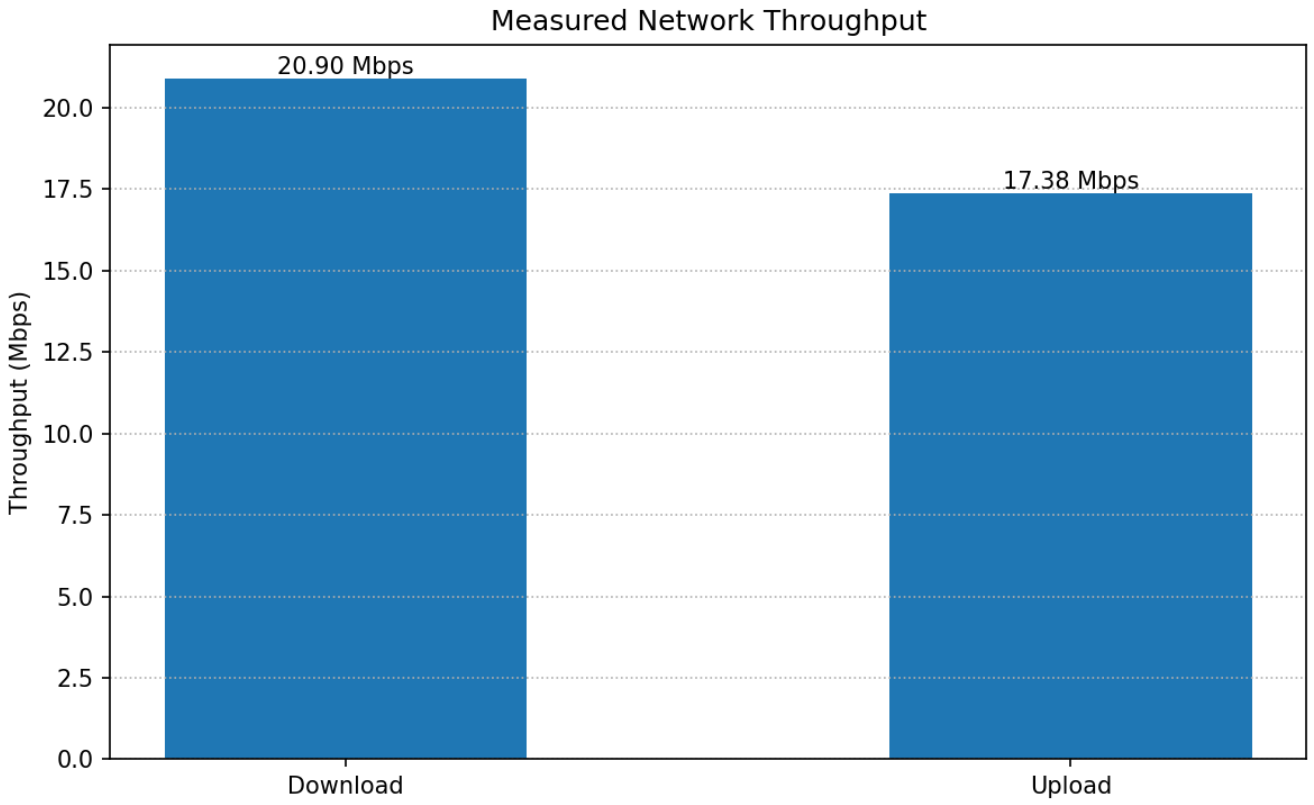
Graph 1: ICMP Round-Trip Time (RTT) Variation



Graph 2: Packet-to-Packet Jitter



Graph 3: Measured Network Throughput


Measured Network Throughput
Throughput (Mbps)
20.90 Mbps
17.38 Mbps
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Download
Upload

EXPERIMENT 8 — Bluetooth Communication & Range Testing — OUTPUT

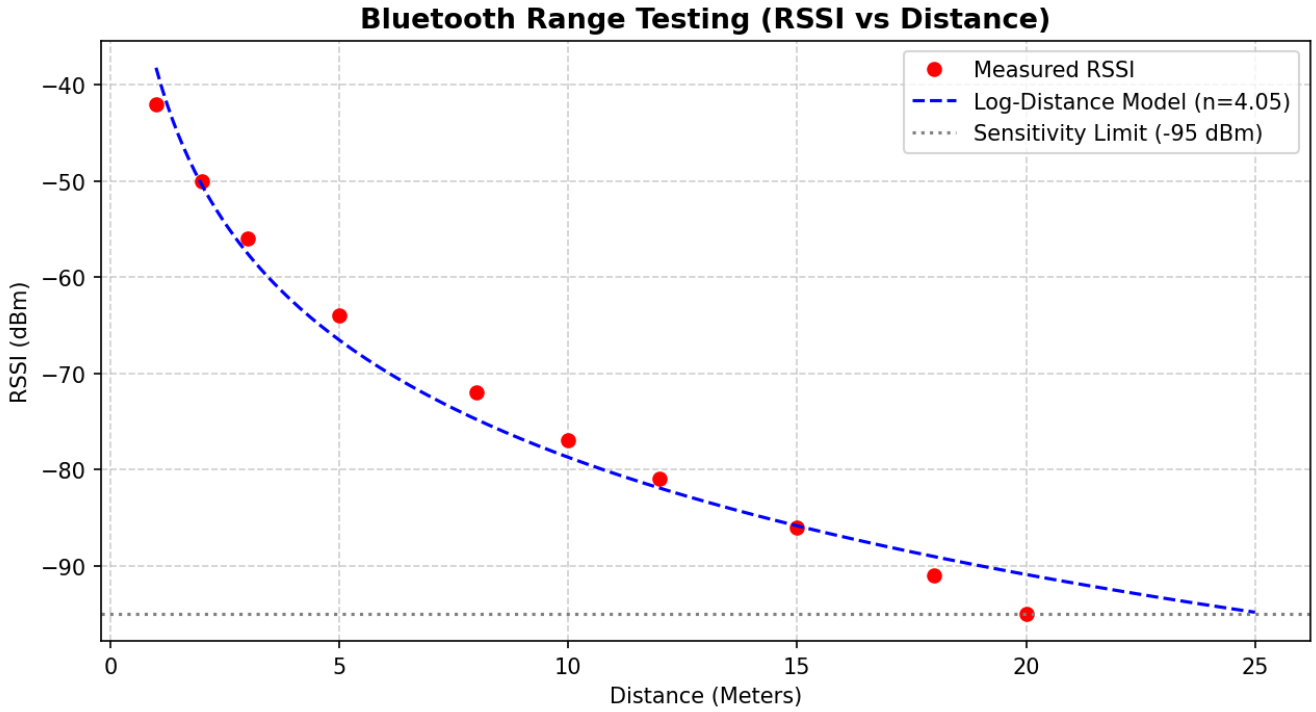
```
Reference Signal at 1m (A)  : -38.21 dBm
Path Loss Exponent (n)      : 4.05
```

RSSI Data:

Distance (m)	RSSI (dBm)
1	-42
2	-50
3	-56
5	-64
8	-72
10	-77
12	-81
15	-86
18	-91
20	-95

Sensitivity Threshold: **−95 dBm**

Graph: Bluetooth Range Testing (RSSI vs Distance)


Bluetooth Range Testing (RSSI vs Distance)
Measured RSSI
Log-Distance Model (n=4.05)
Sensitivity Limit (-95 dBm)
RSSI (dBm)
−40
−50
−60
−70
−80
−90
0
5
10
15
20
25
Distance (Meters)

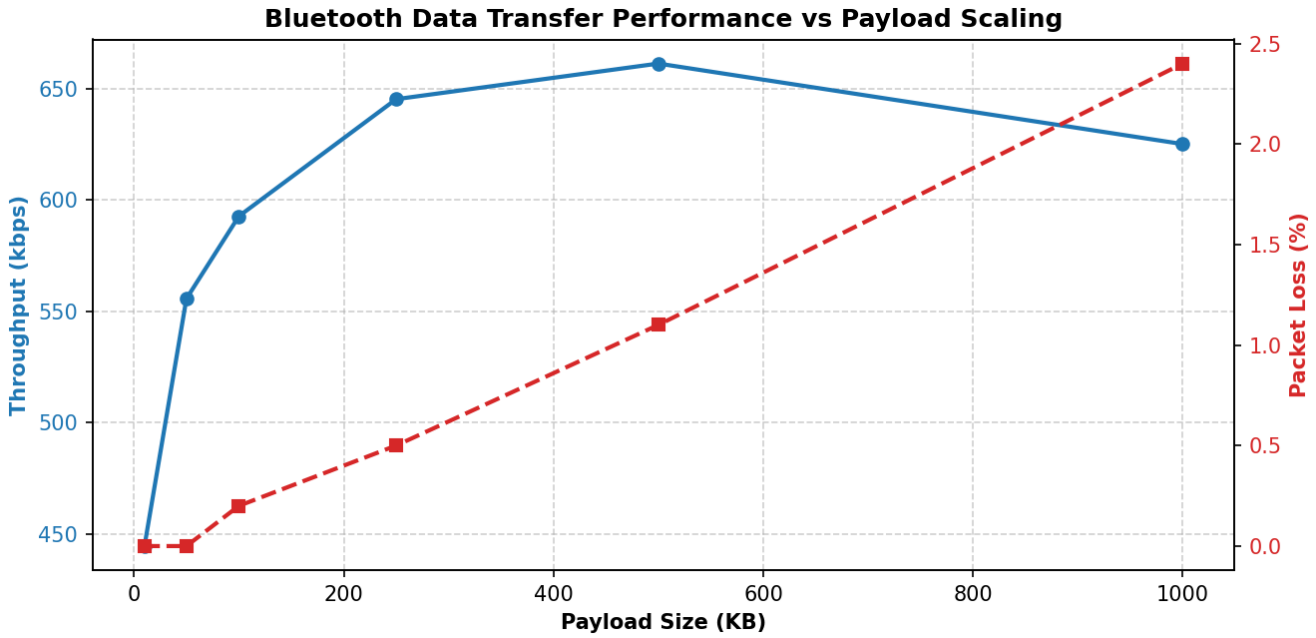
EXPERIMENT 9 — Bluetooth Data Transfer Performance Analysis — OUTPUT

```
============================================================
     EXPERIMENT 9: BLUETOOTH DATA TRANSFER ANALYSIS
============================================================
Payload:   10 KB | Time:  0.18s | Throughput: 444.44 kbps | Loss: 0.0%
Payload:   50 KB | Time:  0.72s | Throughput: 555.56 kbps | Loss: 0.0%
Payload:  100 KB | Time:  1.35s | Throughput: 592.59 kbps | Loss: 0.2%
Payload:  250 KB | Time:  3.10s | Throughput: 645.16 kbps | Loss: 0.5%
Payload:  500 KB | Time:  6.05s | Throughput: 661.16 kbps | Loss: 1.1%
Payload: 1000 KB | Time: 12.80s | Throughput: 625.00 kbps | Loss: 2.4%
```

Observation Table:

Payload Size (KB)	Transfer Time (s)	Calculated Throughput (kbps)	Packet Loss (%)
10	0.18	444.44	0.0
50	0.72	555.56	0.0
100	1.35	592.59	0.2
250	3.10	645.16	0.5
500	6.05	661.16	1.1
1000	12.80	625.00	2.4

Graph: Bluetooth Data Transfer Performance vs Payload Scaling



EXPERIMENT 10 — Indoor Path Loss / Obstacle Analysis — OUTPUT

Obstacle Attenuation Data (at 5 m):

Obstacle	RSSI (dBm)	Attenuation (dB)
Free Space (LOS)	-52	0
Glass Window	-55	3
Wooden Door	-58	6
Single Concrete Wall	-67	15
Double Wall	-78	26

Graph: Indoor Signal Attenuation by Obstacles (at 5m)

